

# Geographic Access to Long-Term Care Facilities for Japan's Aging Population: A Spatial Assessment of Elderly Population Density and Facility Locations in Tokyo

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## ABSTRACT

Japan's aging population presents a major demographic and social challenge, with nearly 30% of citizens aged 65 years or older (Statistics Bureau, 2023). In urban centers such as Tokyo, equitable access to long-term care (LTC) facilities is critical to meet the needs of older adults. This study evaluates the spatial distribution of LTC services relative to elderly populations in Tokyo using ward-level population data and geocoded LTC facility locations (Ministry of Health, Labour, & Welfare, 2022). Geographic Information Systems (GIS) were employed to analyze elderly density per km<sup>2</sup>, general population density, and overlay LTC facility locations with population pyramids. Results indicate that wards with the highest elderly density are generally well-served by LTC facilities; however, peripheral, low-density wards may experience access challenges. Comparative analysis highlights areas of potential inequity, particularly in suburban and mountainous regions. The study demonstrates the value of GIS for identifying spatial disparities in health service provision and for guiding municipal planning decisions. Future research could incorporate facility capacity, mobility patterns, socioeconomic data, and demographic projections to optimize LTC service allocation and accessibility. These findings underscore the importance of aligning health care infrastructure with population needs to maintain the quality of life for Tokyo's aging population (Comas-Herrera et al., 2023; ILO, 2022).

## 1. INTRODUCTION

Japan is experiencing one of the most rapid demographic transitions globally, characterized by low birth rates and increasing longevity (Statistics Bureau, 2023). According to the 2020 census, nearly 30% of the population is aged 65 and older, making Japan the nation with the oldest population structure (Statistics Bureau, 2023). This trend has significant implications for health care delivery, particularly in urban areas such as Tokyo, where dense populations intersect with complex geographic and infrastructural constraints (Health and Global Policy Institute, 2022).

Tokyo, Japan's capital and largest metropolitan area, has a population exceeding 14 million residents and covers approximately 2,194 km<sup>2</sup> (Statistics Bureau, 2023). The city's geography is diverse, including highly urbanized central wards, suburban regions, and mountainous peripheral areas with low population density. This spatial heterogeneity affects accessibility to LTC services, which support older adults who require assistance with daily activities, nursing care, or residential support (Ministry of Health, Labour, & Welfare, 2022).

Japan's LTC system operates under the Long-Term Care Insurance (LTCI) program, providing nursing homes, day-service centers, and other services based on assessed care needs (ILO, 2022). The program has evolved toward a community-based integrated care model, emphasizing coordination of services and equity in access (Morikawa, 2014). While central urban wards often have high facility density, peripheral wards may face challenges due to lower population density and geographic barriers (Ministry of Health,

Labour, & Welfare, 2022; WHO / Kobe Centre, 2020). Understanding the spatial alignment between elderly populations and LTC services is essential for effective resource allocation and policy planning (Comas-Herrera et al., 2023).

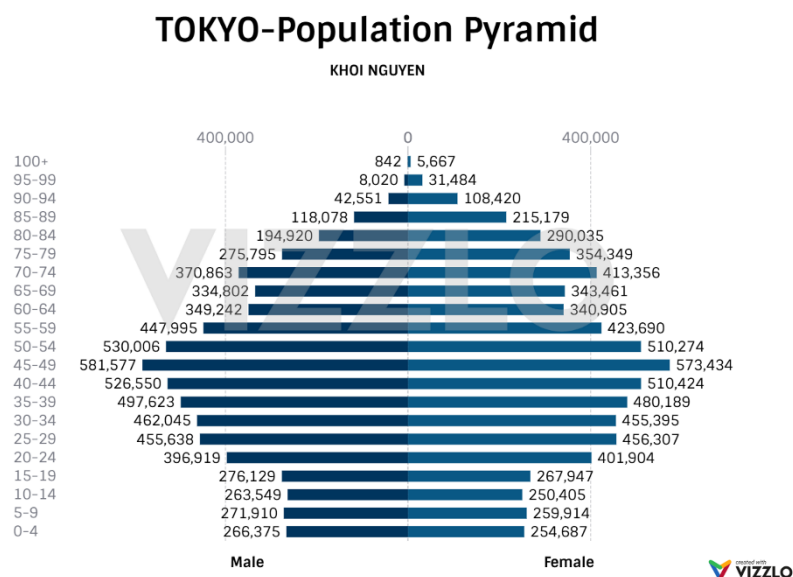
This study evaluates geographic access to LTC facilities in Tokyo by combining ward-level elderly population density, general population density, and facility location data within a GIS framework (Statistics Bureau, 2023). It addresses the following questions:

1. What is the spatial distribution of elderly residents across Tokyo's wards?
2. How does general population density compare to elderly population density?
3. Where are LTC facilities located in relation to population concentrations?
4. Are there wards or regions underserved by LTC services?
5. How can GIS-based analyses inform future planning for LTC services?

## 2. DESCRIPTIVE ANALYSIS

### 2.1 Geography and Population of Tokyo

Figure 1. Tokyo population pyramid, 2020



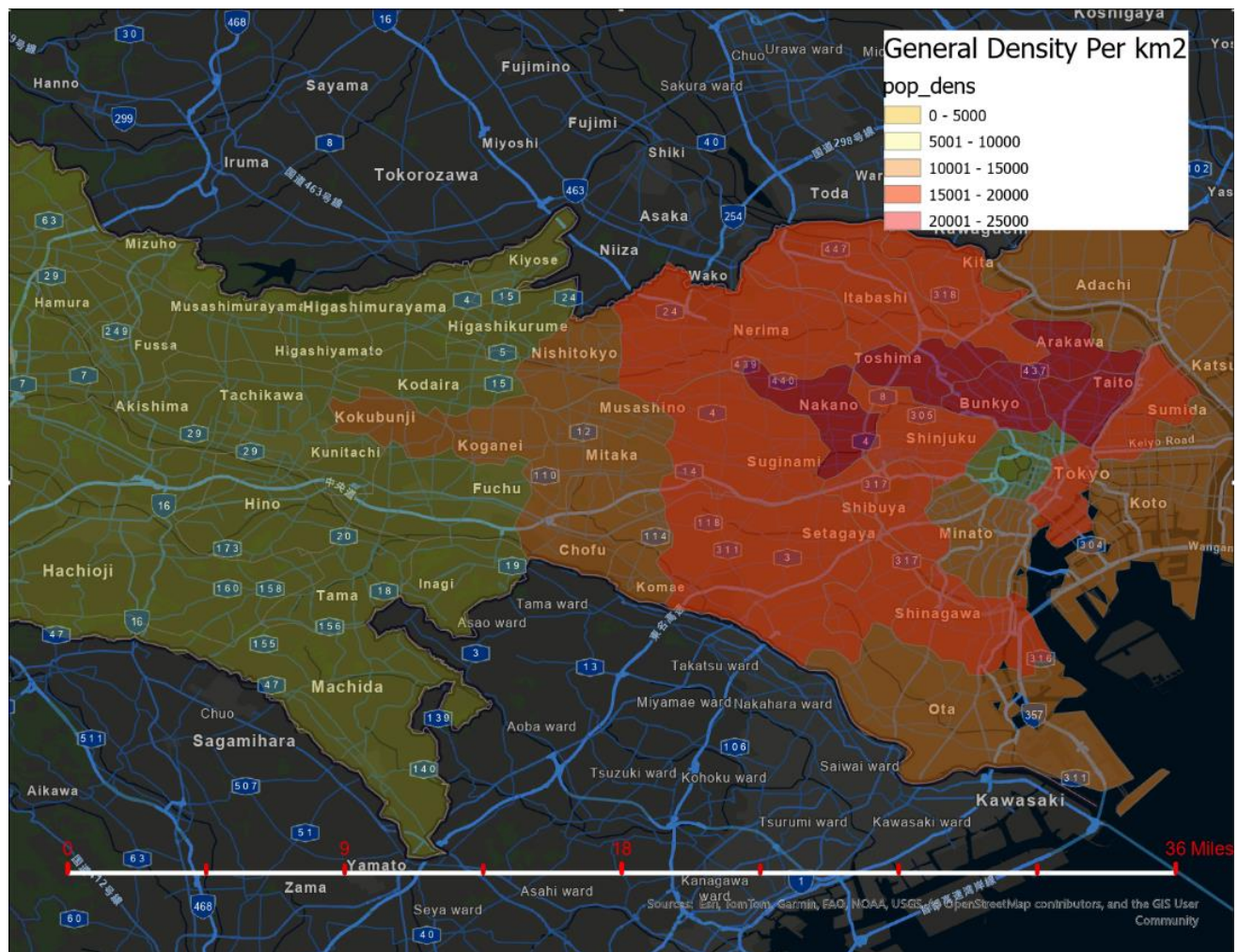
Tokyo is located on the eastern coast of Honshu and consists of 23 special wards in the core urban area, 26 suburban cities, and multiple towns and villages extending into mountainous terrain (Statistics Bureau, 2023). The city's topography ranges from highly urbanized lowlands to sparsely populated mountainous regions, influencing both population distribution and health service accessibility (Health and Global Policy Institute, 2022).

Ward-level demographic analysis shows substantial variation in elderly population density. Central wards, including Shinjuku, Minato, and Chiyoda, exhibit moderate elderly densities due to high overall population, whereas suburban wards such as Setagaya and Ota show high total and elderly population densities (Statistics Bureau, 2023). Peripheral wards, including Okutama, Hinohara, and Ome, have low densities, reflecting rural settlement patterns and geographic barriers (Ministry of Health, Labour, & Welfare, 2022).





Figure 3. General population density per km<sup>2</sup> by ward



## 2.2 Health Care System in Tokyo

Japan's LTCI program provides services to older adults based on age (65+) or disability status, including in-home care, day-service centers, and residential facilities (Ministry of Health, Labour, & Welfare, 2022). The program emphasizes accessibility and equitable service provision; however, geographic disparities remain due to population distribution and city layout (Comas-Herrera et al., 2023).

Central urban wards benefit from high LTC facility concentrations, with multiple services within short travel distances. Peripheral wards face longer travel times and fewer facilities, limiting access for older residents who lack personal mobility or public transit connections (Ministry of Health, Labour, & Welfare, 2022). Municipal health service planning uses demographic projections and current facility distributions, but GIS-based analyses are essential to identify underserved areas (ILO, 2022).

## 2.3 Analytical Framework

The study integrates population and facility data to assess LTC accessibility:

1. Population Analysis: Elderly density per km<sup>2</sup> calculated at the ward level (Statistics Bureau, 2023).
2. Facility Location Mapping: LTC facilities geocoded and mapped to evaluate spatial coverage (Ministry of Health, Labour, & Welfare, 2022).
3. Overlay Analysis: Facilities overlaid on wards with  $\geq 65$  population density to identify alignment and gaps.
4. Accessibility Assessment: Underserved wards identified through spatial overlaps.

Overlay analysis enables the identification of spatial patterns where LTC provision aligns or misaligns with elderly population concentrations (Figures 4–5).

Figure 4. LTC facilities overlaid on  $\geq 65$  population density

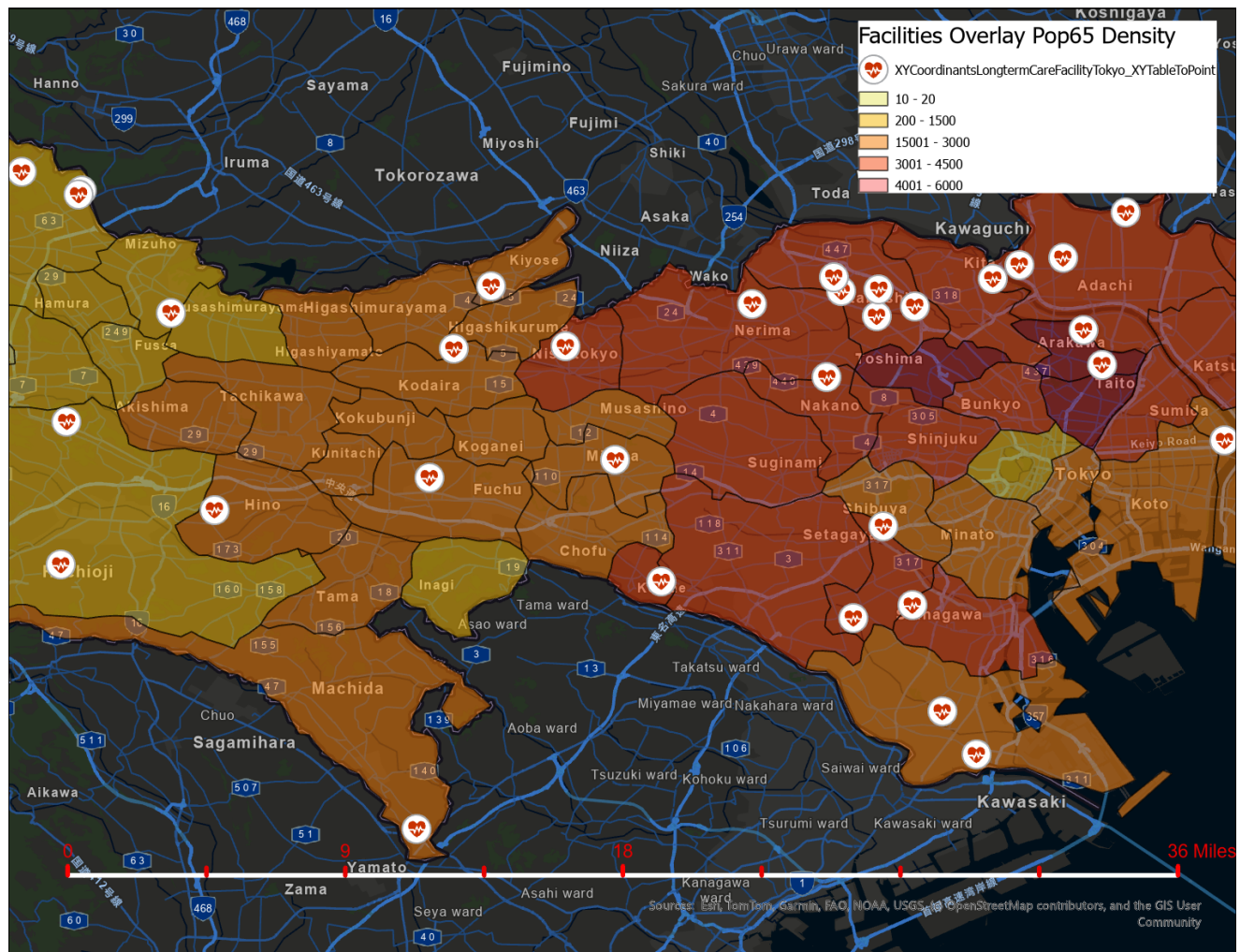
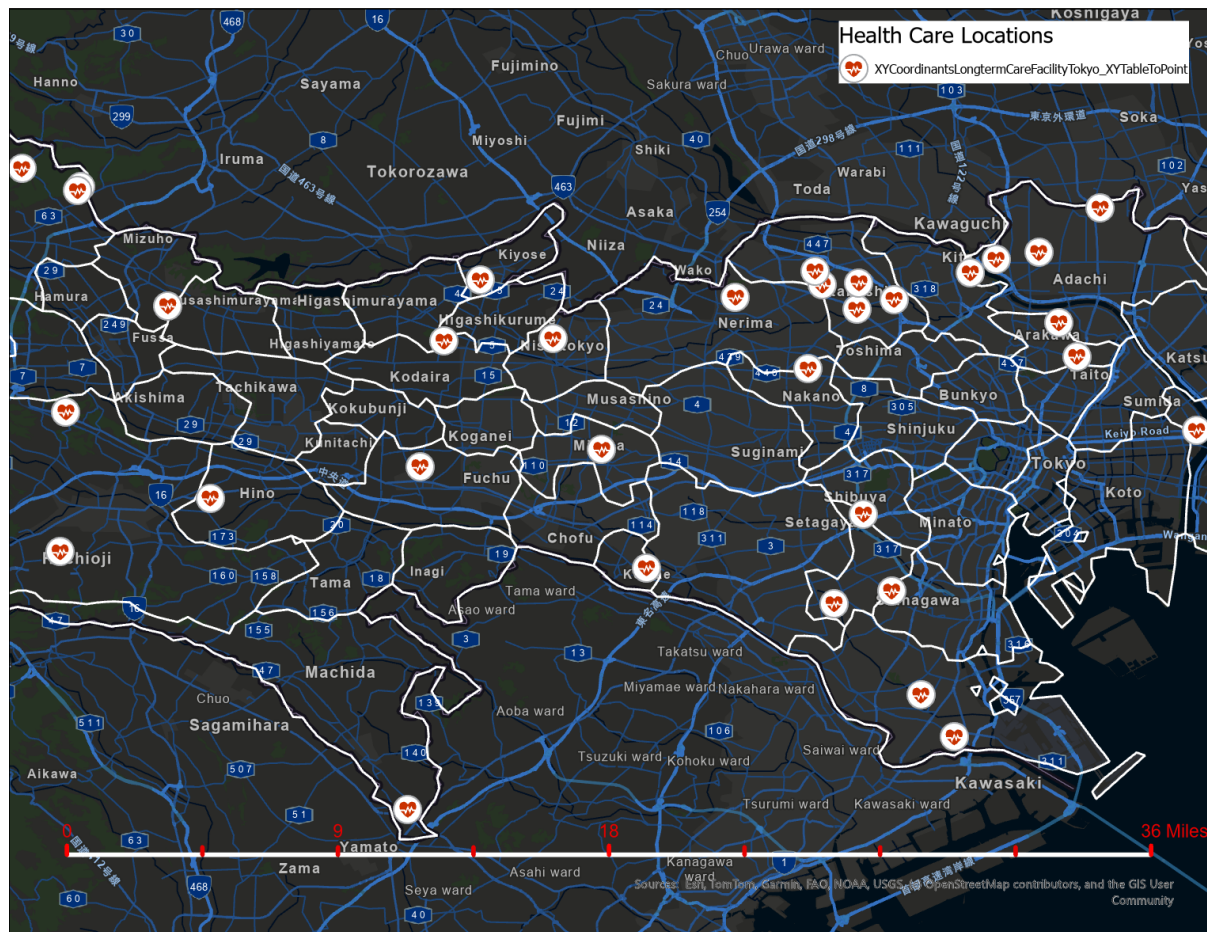




Figure 5. Health care facility locations map



This framework enables identification of spatial patterns, highlighting areas where LTC service provision aligns or misaligns with elderly population concentrations.

## 2.4 Data and Mapping

### Data Sources:

- Population Data: Tokyo ward-level census, 2020 (Statistics Bureau, 2023)
- LTC Facility Locations: Municipal health records and publicly available geocoded lists
- Spatial Boundaries: Tokyo municipal boundary shapefiles

### Data Processing:

- Elderly population densities calculated using total population  $\geq 65$  per ward divided by ward area ( $\text{km}^2$ )
- LTC facility coordinates were obtained by combining publicly available Tokyo government data with a Google Sheets workflow. An Apps Script was used to automatically retrieve latitude and longitude values for each facility address.
- The geocoded LTC facilities were projected in the same coordinate system as ward boundaries.
- Choropleth maps were produced to visualize elderly and general population densities.
- Facility overlay maps were created to assess spatial alignment with elderly populations.
- Many of these processing steps were automated using ArcPy functions, including calculating ward-level

densities, reprojecting shapefiles, and generating maps. This streamlined repetitive tasks and improved reproducibility of the analysis.

Mapping results reveal distinct patterns. Central and suburban wards demonstrate substantial coverage of LTC facilities, whereas peripheral low-density wards are sparsely served. Comparison of elderly versus general population densities (Figures 2–3) demonstrates that high population areas are not necessarily synonymous with high elderly concentration, indicating the need for age-specific planning. This is supported by Comas-Herrera “Comparison of elderly and general population densities demonstrates that total population size alone is not sufficient for LTC planning”.

## 2.5 Comparative Analysis

Overlay analysis of LTC facilities on elderly population density (Figure 4) highlights that many central wards are well-served, with facilities distributed across areas of high elderly concentration (Ministry of Health, Labour and Welfare). However, certain wards, including Tishima, Arakawa, and Taito, show a notable mismatch between elderly population density and facility availability. Despite having relatively high proportions of residents aged 65 and older, these wards have fewer long-term care hospitals and day-service centers compared to central wards such as Shinjuku or Minato (Comas-Herrera et al., 2023).

Comparing general population density to elderly density reveals that overall population size can mislead planners. Wards with moderate total populations but high elderly proportions, such as Arakawa, require more targeted LTC service allocation, while high total population wards with lower elderly concentrations may not require as intensive resources (Statistics Bureau, 2023).

Health care locations mapped independently (Figure 5) provide a complementary perspective, showing clustering in urban centers and highlighting gaps in peripheral areas (Ministry of Health, Labour and Welfare, 2022). These spatial patterns underscore the value of GIS-based analyses for identifying underserved wards and guiding municipal planning decisions to better align LTC resources with population needs (International Labour Organization, 2022).

## 2.6 Summary of Descriptive Analysis

The spatial analysis demonstrates that Tokyo’s LTC service landscape is uneven, reflecting both population density patterns and geographic constraints. Central urban wards show high accessibility, while peripheral and mountainous regions, along with wards like Tishima, Arakawa, and Taito, may face limited access despite high elderly concentrations. The misalignment between elderly population density and LTC facility availability highlights the importance of age-specific analyses for LTC planning and identifies priority areas for policy interventions.

## 3. FURTHER RESEARCH

Building on these findings, future research could pursue several directions to enhance understanding of LTC accessibility and guide planning:

### 3.1 Facility Capacity Analysis

Current analysis assumes all LTC facilities provide equal service capacity. Incorporating bed counts, staffing levels, and service types would allow for more nuanced assessments, especially in wards with high elderly density but limited facilities (Ministry of Health, Labour and Welfare, 2024).

### 3.2 Mobility and Transportation Networks

Accessibility is influenced by more than geographic proximity. Incorporating public transit networks, road travel times, walkability indices, and private vehicle ownership patterns would provide a more realistic measure of LTC access. GIS-based network analyses could simulate travel times from elderly residences to facilities.

### 3.3 Longitudinal Population Projections

With Tokyo's population projected to age further, integrating ward-level population projections would allow planners to anticipate future demand for LTC services and identify high-need areas. Scenario modeling could test the impact of demographic changes on accessibility and equity.

### 3.4 Socioeconomic and Health Indicators

Incorporating variables such as income, household composition, and health status (e.g., prevalence of chronic conditions or disability rates) can help identify populations at greatest risk of LTC inadequacy. Spatial correlations between these indicators and facility locations would support targeted interventions (Economic Research Institute for ASEAN and East Asia, 2022; Morikawa, 2014). Furthermore, integrating policy insights on funding, equity, and long-term care planning can help guide municipal strategies to improve coverage for underserved wards (WHO / Kobe Centre, 2020).

### 3.5 Composite Accessibility Indices

Future research could develop composite metrics that combine elderly density, facility capacity, travel times, and socioeconomic vulnerability into a single index. Such indices could guide evidence-based policy decisions and prioritize interventions.

### 3.6 Advanced GIS Analyses

- Hotspot Analysis: Identify wards with significant clustering of elderly residents relative to LTC facilities.
- Spatial Regression: Model determinants of LTC facility placement and accessibility.
- Scenario Modeling: Simulate expansion or relocation strategies and evaluate effects on accessibility metrics.

These approaches would improve understanding of spatial inequities in LTC provision and inform policy strategies to optimize coverage.

## 4. CONCLUSIONS

This study demonstrates that geographic analysis is essential for evaluating LTC access in Tokyo. Central urban wards exhibit higher coverage relative to elderly density, whereas peripheral and low-density wards, including Tishima, Arakawa, and Taito, face access challenges. Overlay analyses and choropleth mapping reveal areas where LTC provision is misaligned with population need.

GIS-based methods allow planners to visualize spatial patterns, compare demographic distributions, and identify underserved populations. Integrating additional data on facility capacity, mobility, socioeconomic indicators, and population projections would further strengthen planning decisions. Aligning LTC services with the spatial distribution of elderly residents is critical to maintaining equitable access and quality of life for Tokyo's aging population.



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